Curriculum Vitae

Personal Information

Last name	Zodji
First name	Marcel
Date of birth	April 6, 1997, in Houin, Benin
Civil status	Father of a daughter born in June 2026
Website	https://mzodji.github.io/home
E-mail	marcel.zodji@univ-evry.fr

Experience

Sept. 2025 – Aug. 2027	**Postdoctoral Researcher** at LaMME, Université d'Évry-Val-d'Essonne. **Supervisors**: Thomas Alazard and Mickaël Latocca. **Funding**: Fondation Mathématique Jacques Hadamard (FMJH).
Sept. 2025 –	**Teaching Assistant** at the Université d'Évry-Val-d'Essonne. Real Analysis 1 (L1 – Math-Physique – 36 hours) Mathematical Language (L1 – Math-Info – 18 hours)
2024 – 2025	**Teaching and Research Assistant** (ATER) at Université Paris Cité – 192 hours. Elementary Mathematics (L1 – Physics), Mathematics for Biologists (L2 – Biology), Discrete Probability (L2 – MIASHS), Mathematics for Engineers (L2 – CUPGE).
2021 – 2024	**Research and Teaching Assistant** at Université Paris Cité – 3×64 hours. Mathematics and Applications to Life Sciences (L1 – PASS), Discrete Probability (L2 – MIASHS), Office and Internet Tools (L1 – Math-Info).
Aug. 2019	**Internship** at the National Institute of Statistics and Economic Analysis (INSAE), Benin.

Education

2021 – 2025	**PhD** in Mathematics at Université Paris Cité **Thesis**: Qualitative study of the interface between two immiscible compressible fluids. **Defense date**: June 17, 2025. **Supervisors**: Cosmin Burtea and David Gérard-Varet.
2020 – 2021	**Master's degree** in Mathematical Modelling at Sorbonne Université, France.
2018 – 2020	**Master's degree** in Fundamental Mathematics and Applications at the Institut de Mathématiques et de Sciences Physiques (IMSP), Benin.
2017 – 2018	**Bachelor's degree** in Mathematics at IMSP, Benin.
2015 – 2017	**Preparatory classes for the Grandes Écoles** at IMSP, Benin.
2015	**Baccalauréat**, Série C, Benin.

Grants and Awards

2025	Postdoctoral thematic fellowship from FMJH.
2021	Doctoral fellow in the MathInParis 2020 Cofund programme of the Fondation Sciences Mathématiques de Paris (FSMP).
2020	Master's fellow in the PGSM programme of FSMP, co-funded by the French Embassy in Benin.
2018	Recipient of a scholarship from the German Academic Exchange Service (DAAD) for master's studies in Benin.

Responsibilities

Feb. 12, 2026	**Club mademoiselle maths**. Talk on my career path and my interest in mathematics, given to middle-school students (girls) visiting LaMME.
2022 –	**Volunteer at ZUPdeCO** Online mathematics tutoring for high-school students.
2022 – 2023	**Mentor** for the PGSM fellows program of FSMP.
Oct. 2021	**Fête de la Science**.

Research

My thesis work focuses on free-boundary problems for models describing the flow of fluid mixtures. The components are assumed to be immiscible (for example, gas bubbles in a liquid) and separated by one or more sharp interfaces. The goal is to study the propagation of interface regularity over time, including the well-posedness of the models. We use Hoff-type energy estimates combined with Lagrangian changes of variables and various tools from harmonic analysis. This work has resulted in three published journal articles, one preprint, and a submitted manuscript.

1. **Global-in-time well-posedness of the compressible Navier-Stokes equations with striated density**. With Xian Liao.
   Revista Matemática Iberoamericana 41 (2025), no. 6, pp. 2167–2214.
   https://doi.org/10.4171/rmi/1561.
2. **Discontinuous solutions for the Navier-Stokes equations with density-dependent viscosity**.
   Ann. Inst. H. Poincaré C Anal. Non Linéaire (2024), vol. 43, no. 1, pp.1-58.
   https://doi.org/10.4171/aihpc/145.
3. **A well-posedness result for the compressible two-fluid model with density-dependent viscosity**.
   Journal of Mathematical Fluid Mechanics, vol. 27, no. 50, 2025.
   https://doi.org/10.1007/s00021-025-00954-y.
4. **Global-in-time discontinuous solutions for the two-phase model of compressible fluids with density-dependent viscosity**.
   Preprint. https://arxiv.org/abs/2510.11383 or https://hal.science/hal-05312124.
5. **Global-in-time well-posedness of the two-dimensional compressible Navier-Stokes equations with general density-dependent viscosity coefficients**.
   With Xian Liao and Rebekka Zimmermann. 2026 – Submitted – Links: coming soon.

Research Visits

2023–2025	**Karlsruhe Institute of Technology**, Germany. Invited by **Xian Liao** for one-week research visits each July.
Oct. – Dec. 2023	**Karlsruhe Institute of Technology**, Germany. **Host**: Xian Liao **Funding**: CRISIS project (ANR-20-CE40-0020-01) **PI**: Francesco Fanelli

Scientific Talks

Nov. 24–26, 2026	**Journées des Jeunes EDPistes**. Bordeaux, France.
Oct. 20, 2026	**Séminaire de l'équipe Physique Mathématiques et EDP**. LAGA, Institut Galilée, Université Sorbonne Paris Nord.
Sept. 14, 2026	**Séminaire Analyse et EDP**. Laboratoire AGM, Cergy, France.
Nov. 12, 2025	**Online talk**. School of Mathematical Sciences, Beijing Normal University.
Sept. 21–26, 2025	**MathFlows'25 Workshop**. Île de Porquerolles, France.
Apr. 15, 2025	**Séminaire MAC**. Institut de Mathématiques de Toulouse.
Mar. 20, 2025	**Séminaire de l'équipe d'Analyse et EDP**. LaMME, Université d'Évry-Val-d'Essonne.
Sept. 2–4, 2024	**Meeting on Differential Equations in Ancona**. Ancona, Italy.
Feb. 23, 2024	**Séminaire EDPs2**. LAMA, Université Savoie Mont Blanc, Chambéry, France.
Jan. 31, 2024	**Séminaire des doctorant.es du LJLL**. LJLL, Paris, France.
Nov. 16, 2023	**Conférence "Maths pour la Géophysique"**. Amiens, France.
July 25, 2023	**Working group "Nonlinear PDE"**. Karlsruhe Institute of Technology, Germany.

Scientific Poster Presentations

Nov. 30–Dec. 5, 2026	**Mathematical Models for Mixtures**. CIRM, Marseille, France.
Oct. 27–31, 2025	**Workshop "End of CRISIS"**. CIRM, Marseille, France.
June 23–27, 2025	**Summer School**, Particles in Flow. Bonn, Germany.
Jan. 8–9, 2024	**Journées de lancement de l'ANR BOURGEONS**. LJLL, Paris, France.
Mar. 22, 2023	**Journées des Jeunes EDPistes**. Tours, France.

Reviewing Activities

Zeitschrift für Angewandte Mathematik und Physik (ZAMP)
Applied Mathematics Letters

Skills

Languages	French (good), English (intermediate)
Computing	Python, Scilab, C, LaTeX.

References

Thomas Alazard	CNRS Research Director, École Polytechnique, thomas.alazard@polytechnique.edu.
Cosmin Burtea	Associate Professor, Université Paris Cité, cosmin.burtea@imj-prg.fr.
David Gérard-Varet	Professor, Université Paris Cité, david.gerard-varet@u-paris.fr.
Mickaël Latocca	Associate Professor, Université d'Évry-Val-d'Essonne, mickael.latocca@univ-evry.fr.
Xian Liao	Junior Professor, Dalian University of Technology (China), liao@dlut.edu.cn.